

Exploring Multi-Order Markov Chains for Road Network Mobility Prediction

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Abstract

Accurate mobility prediction is essential for intelligent transportation systems, yet existing methods are either too complex for real-time deployment or lack online adaptability. We study a family of multi-order Markov chain models that progressively enrich how trajectory history is used: from selecting the longest matching context, to combining information across multiple history lengths, and finally to integrating population-level patterns with online individual-level memory. Under a dual evaluation protocol combining next-node prediction and auto-regressive forecasting, the proposed lightweight models match or outperform established baselines in predictive accuracy. The results show that higher-order memory improves short-term prediction and long-horizon consistency, while the dual-memory variant reduces predictive uncertainty and improves endpoint accuracy. In addition, we adapt WorldMove, a recent synthetic mobility dataset, to road-network graphs, providing an open benchmark for next-node prediction, graph-based trajectory forecasting, and broader studies of mobility over transportation networks.

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1 Introduction

Understanding and anticipating human movement has emerged as a critical problem covering a wide range of real-world applications, from intelligent transportation systems and traffic management to personalised navigation and urban planning [4, 2, 15], making the accurate prediction of the next node in a road network trajectory a cornerstone of modern mobility modelling. Recent deep learning approaches [1, 9, 14] achieve strong next-node prediction performance through long-range temporal modelling and rich representation learning. However, these models require full offline training pipelines, substantial computational resources, and offer no lightweight mechanism for adapting to an ongoing trajectory in real time, properties that are critical for deployment in real-time or distributed mobility systems. Higher and variable-order sequence models (e.g. AKOM [12], CPT+ [7], MOGen [5], and de Bruijn graphs [13, 16]) reduce this computational load but remain sensitive to order selection. This raises a natural question: can simpler, more transparent models achieve competitive performance while remaining lightweight enough for real-time and distributed use? To investigate this, we explore a family of Markov Chain models of increasing complexity. Next-node prediction algorithms can be broadly classified into *individual* methods, which exploit single-user history to capture personal mobility patterns, and *collective* methods, which leverage population-level regularities to generalise across users [8]. The former suffer from cold-start



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when personal history is limited; the latter often lack an online adaptation mechanism to personalise predictions. Understanding which of these components is necessary or sufficient for accurate prediction is precisely what a transparent, modular model family allows us to uncover. Here, we evaluate three Markov Chain variants of increasing complexity: a simple maximum order model (MC), an entropy-weighted variant (HW-MC), and a Dual-Memory Markov Chain (DM-MC), which combines a *Population-level Memory* (PM) built offline from a set of training trips with an *Individual-level Memory* (IM) updated online with the user’s trajectory history. To evaluate these models, we introduce a dual evaluation protocol on road networks derived from WorldMove [15]. First, we consider next-node prediction, where ground-truth observations remain available throughout the sequence. Second, we evaluate auto-regressive forecasting, which reflects settings where movement is only intermittently observed on the road network: from a known starting location, the model must predict the next t steps until the next sensor or tracking update becomes available. A key contribution of our setup is that we adapt WorldMove, originally a synthetic mobility dataset, to *OpenStreetMap*-based road graphs. This turns WorldMove into a benchmark for studying mobility directly on road topology, enabling further experiments in network-based trajectory modelling.

2 Markov Chain Models for Mobility Prediction

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a road network graph. Given a partial trajectory $(v_1, \dots, v_t) \in \mathcal{V}^t$, the objective is to predict the next node v_{t+1} . We consider three progressively richer Markov Chain models that address the possible limitations of their predecessors.

Max-Order Markov Chain (MC): The first model is a fixed maximum-order Markov Chain of order K . For a context $c^{(k)} = (v_{t-k+1}, \dots, v_t)$, next-node distributions are estimated from k -gram transition counts $C_{c,v}^{(k)}$ accumulated over training trajectories, with support $N_c^{(k)} = \sum_{v'} C_{c,v'}^{(k)}$. At prediction time, the model selects the highest order whose context was observed and returns its empirical distribution

$$P_{MC}(v) = \frac{C_{c,v}^{(k^*)}}{N_c^{(k^*)}}. \quad (1)$$

While simple and effective, using a single order requires trading contextual specificity against statistical reliability.

Entropy-Weighted Markov Chain (HW-MC): To alleviate this trade-off, HW-MC retains all observed orders $k \in \{1, \dots, K\}$ and combines their empirical distributions $P^{(k)}(v) = \frac{C_{c,v}^{(k)}}{N_c^{(k)}}$. Orders with lower entropy contribute more strongly, while an evidence-discounting factor reduces the influence of poorly supported contexts:

$$P_{HW-MC}(v) = \sum_k \alpha^{(k)} P^{(k)}(v), \quad \alpha^{(k)} \propto \left(\frac{N_c^{(k)}}{N_c^{(k)} + \tau} \right)^\lambda (H(P^{(k)}) + \varepsilon)^{-1}, \quad (2)$$

where $H(P) = -\sum_v P(v) \log P(v)$, $\tau > 0$ is the support saturation scale, $\lambda \geq 0$ the discount exponent, and $\varepsilon > 0$ prevents division by zero. If no order has support, the model falls back to the unigram distribution and finally to a uniform distribution. HW-MC therefore isolates the effect of adaptive multi-order aggregation.

Dual-Memory Markov Chain (DM-MC): Although HW-MC exploits multiple orders, it relies exclusively on population-level patterns. DM-MC augments this *Population Memory* (PM) with an *Individual Memory* (IM) that captures recurrent user-specific mobility behaviour. The PM is obtained by aggregating per-order distributions using the HW-MC weights $\alpha^{(k)}$, yielding P_{PM} . The IM scans the user’s history buffer (the last N visited nodes across previous trips) for occurrences of the query context $c^{(k)}$, recording successor counts $n^{(k)}(v)$ and applying symmetric Dirichlet smoothing:

$$P_{\text{IM}}^{(k)}(v) = \frac{n^{(k)}(v) + \beta/V}{n^{(k)} + \beta}. \quad (3)$$

Per-order IM distributions are aggregated similarly to PM, and unmatched orders back off to a uniform distribution. The two memories are then combined as

$$P_{\text{DM-MC}}(v) = \frac{w_{\text{PM}}P_{\text{PM}}(v) + w_{\text{IM}}P_{\text{IM}}(v)}{w_{\text{PM}} + w_{\text{IM}}}, \quad w_{\text{PM}} = (H(P_{\text{PM}}) + \varepsilon)^{-1},$$

$$w_{\text{IM}} = (H(P_{\text{IM}}) + \varepsilon)^{-1}(1 + \eta\rho), \quad \rho = \frac{N_{\text{IM}}}{N_{\text{IM}} + \tau} \left(\frac{k_{\text{max}}}{K} \right)^p \left(\max_v P_{\text{IM}}(v) \right)^\kappa. \quad (4)$$

where N_{IM} is the total number of context matches in the user’s history, k_{max} the highest matched order, $\eta \geq 0$ the personalization strength, and p, κ control the influence of order and confidence. As user-specific evidence accumulates, the model progressively shifts from population-level predictions toward personalized behaviour.

3 Methodology

We evaluate the proposed Markov Chain variants against four established sequential prediction models: AKOM [12], which predicts the next element by matching the longest suffix of the current sequence against a variable-order context tree; CPT+ [7], which stores training sequences in a compact trie and retrieves predictions via similarity-based lookup; MOGen [5], which models higher-order transition dynamics through a multi-order graph representation; and IOHMM [11], a latent-state model that captures individual mobility regularities from user-specific history. We reimplement AKOM, CPT+, and MOGen to ensure a fair comparison, and use the publicly available implementation for IOHMM. All models are evaluated on a common dual experimental setup under 5-fold cross-validation over four road networks of varying scale and topology, with trips drawn from WorldMove [15], a synthetic mobility dataset that mimics the statistical properties of daily human movement. WorldMove generates trajectories by combining publicly available geographic data with a diffusion-based generative model, producing spatiotemporal sequences of visited locations over time. For each city, we retrieve the corresponding road network from *OpenStreetMap* (OSM) and map each trajectory onto it, so that every trip becomes a sequence of OSM node identifiers. Statistics of the resulting datasets are summarised in Table 1.

Table 1 Statistics of the four WorldMove datasets converted into OSM graphs format.

Dataset	$ \mathcal{V} $	$ \mathcal{E} $	Trips	Users	Days	Med. dist (km)	Med. dur (min)	Med. path nodes
Abu Dhabi (AE)	53,251	112,591	18,676	500	15	16.01	32.02	81
Frankfurt (DE)	59,219	139,436	34,148	500	15	11.47	22.93	89
New York (US)	128,785	331,874	142,383	500	15	12.45	24.89	107
Singapore (SG)	43,941	91,939	76,421	500	15	12.02	24.04	86

The dual evaluation protocol comprises two complementary tasks, each reflecting a distinct operational context. **Next-node prediction** assumes continuous access to ground-truth location data: given a context window of the previous 10 nodes $X_t = (v_{t-9}, \dots, v_t)$,

the model predicts the immediately following node v_{t+1} , and all possible sliding windows are extracted from each test trajectory to form context–target pairs. **Auto-regressive trajectory forecasting** addresses a stricter mode of operation: given only an initial seed of the first 10 ground-truth nodes, the model generates a full predicted trajectory without any further real-time input. At each step, the model’s own prediction is fed back as context, $X_{t+1} = (v_{t-8}, \dots, v_t, \hat{v}_{t+1})$, where $\hat{v}_{t+1}$ denotes the predicted node, so that errors may accumulate over time, making this a strictly harder evaluation task. All models and datasets used in this work are publicly available through the accompanying repository [3]. For MC, HW-MC, and DM-MC, experiments are conducted across three orders ($k \in \{2, 5, 10\}$); the same order configurations are applied to AKOM, CPT+, and MOGen where applicable. For DM-MC, the Individual Memory (IM) is constructed from the 3000 most recent nodes drawn from the user’s held-out history trips. For the next-node task, performance is measured using: the *Accuracy*, the proportion of samples for which the highest-ranked prediction is the true next-node; and the *Negative Log-Likelihood* (NLL), defined as $NLL = \frac{1}{N} \sum_{i=1}^N -\log_2 P(y_i | X_t^{(i)})$, expressed in bits and measuring *uncertainty*. A lower NLL means the model concentrates more probability mass on the node that actually occurs, i.e. it is both correct *and* certain; two models with identical *Accuracy* can differ in NLL, with the lower-NLL model being better calibrated and more decisive. For the auto-regressive task, performance is measured using: the *Acceptable Node Horizon Prediction*, the mean number of consecutive future nodes the model can predict in closed-loop rollout while its cumulative top-1 accuracy stays $\geq 75\%$; and the *Endpoint distance*, the distance between the final node of the predicted path and the final node of the ground-truth.

4 Results

Results show that accuracy improves consistently with Markov order across all models, with higher-order contexts providing stronger predictive signal (see Table 2). For the next-node task, MC, HW-MC, and DM-MC reach comparable accuracy at equivalent orders, with MC10 achieving the highest overall accuracy (0.9490) and DM-MC10 and HW-MC10 close behind (0.9477 and 0.9486 respectively, demonstrating that these simple, lightweight Markov Chain variants match or outperform more complex established models such as CPT+ (0.9439) and IOHMM (0.9329) on this task. The advantage of DM-MC, however, lies in predictive confidence: despite near-identical *Accuracy*, DM-MC achieves dramatically lower *NLL* than all competing models. DM-MC10 reaches 0.29 bits against 10.73 for MC10 and 2.31 for HW-MC10, an order-of-magnitude gap that holds across all orders, indicating that DM-MC not only ranks the correct node highly, but concentrates substantially more probability mass on it, producing predictions that are both accurate and decisive, whereas baselines remain uncertain even when correct. The auto-regressive task is substantially harder: error accumulation along the predicted path causes all models to converge toward similarly modest performance. MC10 achieves the highest Node Horizon Prediction (18.06 nodes), ahead of HW-MC10 (15.31) and MOGEN (15.60), while DM-MC10 reaches 14.82 nodes, competitive with CPT and HW-MC5 but inferior to the plain Markov baselines, suggesting that the dual-memory mechanism, while beneficial for confidence, does not provide additional benefit for long-horizon rollout over strong higher-order chains. DM-MC10 does however achieve the lowest endpoint distance (5.03 km) among all models, indicating that the trajectories terminate spatially closer to the ground truth even when intermediate predictions diverge. Overall, performance differences across models remain modest, reflecting a structural ceiling imposed by road-network out-degree: once errors compound over several steps, local

Table 2 Prediction accuracy, certainty, reliable look-ahead depth, and endpoint distance, averaged over 4 datasets (and standard deviation).

Model	Next-node task		Auto-regressive task	
	Accuracy $\uparrow$	NLL (bits) $\downarrow$	Node horizon pred. (75%) $\uparrow$	Endpoint dist. (km) $\downarrow$
DM-MC2	0.9315 (0.0124)	0.3947 (0.0923)	13.26 (1.43)	7.2543 (2.5482)
DM-MC5	0.9437 (0.0144)	0.3052 (0.0976)	14.62 (1.96)	6.7841 (2.6898)
DM-MC10	0.9477 (0.0148)	0.2924 (0.1058)	14.82 (1.61)	5.0340 (1.4205)
AKOM	0.8844 (0.0519)	1.3130 (0.3915)	13.23 (1.29)	5.2928 (1.3686)
CPT	0.9439 (0.0161)	1.5978 (1.2182)	14.15 (1.67)	6.2676 (1.7758)
IOHMM	0.9329 (0.0119)	2.6111 (2.2113)	12.96 (1.33)	7.3373 (1.7883)
MOGEN	0.9142 (0.0497)	1.1285 (0.9412)	15.60 (7.01)	5.3783 (1.4069)
HW-MC2	0.9319 (0.0123)	3.4531 (3.3524)	13.43 (1.47)	6.5730 (1.5765)
HW-MC5	0.9435 (0.0137)	2.2562 (2.3922)	14.03 (1.70)	6.9654 (1.1441)
HW-MC10	0.9486 (0.0136)	2.3122 (2.4465)	15.31 (1.70)	6.1721 (2.0880)
MC2	0.9319 (0.0123)	3.4531 (3.3524)	13.43 (1.47)	6.5730 (1.5765)
MC5	0.9442 (0.0139)	5.8973 (5.3674)	17.23 (6.20)	6.3507 (1.7695)
MC10	0.9490 (0.0130)	10.7262 (8.9439)	18.06 (5.70)	5.9047 (1.9545)

topology constrains reachable nodes and limits the discriminative power of the task. Beyond model comparison, however, these results carry a practical implication: the auto-regressive setting reveals how many steps t a model can forecast offline before accuracy degrades to an uninformative level, directly informing the maximum admissible interval between consecutive sensor or tracking updates in real-world deployments.

5 Conclusion & Future Work

We explored a family of multi-order Markov Chain models of increasing complexity, progressively combining adaptive order selection with dual population-individual memory. We propose a dual evaluation protocol contrasting next-node prediction, with continuous ground-truth updates, against auto-regressive forecasting, where trajectories are generated from a seed without real-time input. We also proposed graph-mapped versions of the WorldMove mobility datasets, enabling future research that explicitly accounts for road topology. Under this protocol, MC10 achieves the highest next-node accuracy and the deepest node horizon in auto-regressive forecasting. DM-MC10 approaches this accuracy while assigning substantially lower uncertainty to predicted nodes and achieves the lowest endpoint distance, confirming that the dual-memory architecture preserves long-horizon forecasting ability while gaining precision on both uncertainty quantification and destination prediction. Notably, all three Markov Chain variants match or outperform more complex established baselines on next-node accuracy, reinforcing that lightweight, interpretable models are not merely sufficient but competitive, delivering comparable or superior performance without the computational overhead of heavier sequential prediction models. Future work will focus on learning cluster-specific PMs to capture population heterogeneity, evaluating sensitivity to varying context window lengths, and exploring alternative node encoding schemes beyond the categorical identifiers, including higher-order *de Bruijn* encodings and graph embeddings such as *Node2Vec* [6], to assess whether richer structural representations of the road network further improve prediction. We also plan automatic tuning of parameters with *irace* [10] and a more detailed PM/IM contribution analysis via regret-based measures [8].

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